



Sultan Qaboos University

College of Engineering

Petroleum and Chemical Engineering Department

Chemical Reaction Engineering

CHPE4512

Production of Cumene

Memo 3

Section 02

Group 03

Submitted by:	ID	Task
Mohammed Ibrahim Al-Hinai	138348	Part 1,3
Omar Sulaiman Al-Ruqaishi	141761	Leader (Part 1,4)
Majid Mohammed Al-Nabhani	138708	Part 2,3
Omar Yahya Al-Shaqsi	141955	Part 1,4
Sufiyan Humaid Al-Fazari	139364	Part 2,3
Husam Jaber Al-Rawahi	139369	Part 1

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Chapter 3: Multi Tabular Fixed Bed Reactor Design with Pressure Drop Analysis

3.1 Preliminary Single Pipe Reactor Design

In this section, a preliminary design of a single-pipe fixed-bed reactor is proposed, based on the ideal isothermal model but extending it to include a momentum balance, reflecting the actual phenomenon of pressure drop in the reactor. In gas-phase reactions, the pressure drop constraint must not exceed one-tenth of the initial inlet pressure, which can be achieved by carrying out a mole balance and the Ergun equation, which represents the momentum balance.

The model is based on several assumptions to make the computation process simpler:

- The reactor operates in isothermal and steady-state conditions.
- Only one primary reaction occurs in the reactor, and no side reactions are involved.
- The model begins with a single pipe structure before scaling.

Thus, by using this balance equation in conjunction with the Ergun equation, it is possible to obtain a calculation of the change in pressure, or dy/dw , in addition to the conversion. This provides an opportunity to examine the effect of frictional losses in the catalyst, based upon particle size and void fraction, upon the partial pressures of the reactants and, in turn, upon the rate of reaction and weight of catalyst required.

For Calculation part the Ergun equation is used to determine pressure drop in packed bed reactors:

$$\frac{dp}{dz} = - \frac{G}{\rho g_c D_p} \left(\frac{1 - \phi}{\phi^3} \right) \left[\frac{150 (1 - \phi) \mu}{D_p} + 1.75 G \right]$$

And for the calculation, the equation below is derived and used:

$$\frac{dp}{dW} = \frac{-\beta_0}{2y \times A_c \times (1 - \phi) \rho_c} (1 + \epsilon x)$$

$$\beta_0 = \frac{G(1 - \phi)}{g_c \rho_0 D_p \phi^3}$$

$$\alpha = \frac{2\beta_0}{A_c (1 - \phi) \rho_c \rho_0}$$

The Effect Catalyst Weight on Conversion

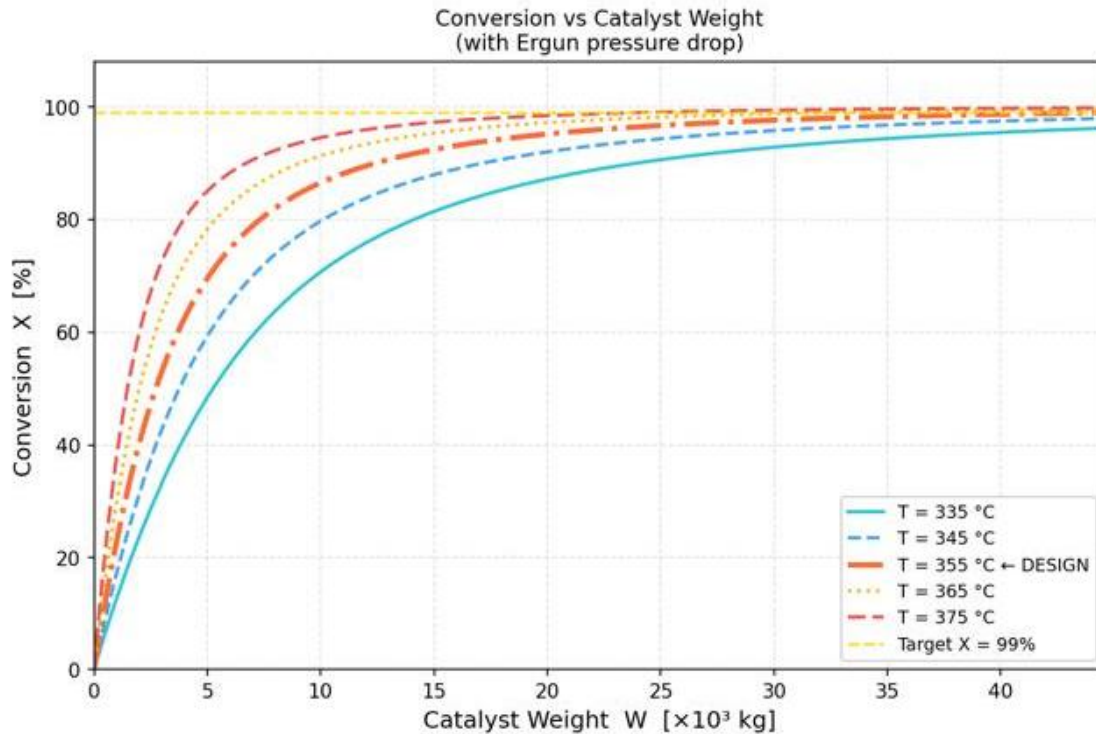


Figure 1: Conversion Vs Catalyst weight

The above plot shows the relationship between the fractional conversion of the reactant material (X, given in percentage on the y-axis) and the weight of the catalyst material it passes through (W, given in units of 10^3 kg on the x-axis). The data points are modeled for a single pipe reactor with a bed depth-to-diameter ratio of 5.00 and incorporate momentum loss by the Ergun equation.

Key Observations:

- **Temperature Dependence:** The graph shows that there are five unique curves, indicating that there are various isothermal operating temperatures, ranging from 335°C to 375°C. As predicted by the Arrhenius equation, increasing the reaction temperature accelerates the reaction rate.
- **High Temperature (375°C - Red dashed line):** At the peak temperature, the reaction is extremely fast. The curve is extremely steep at the start and reaches the 99% conversion point at the lowest weight of catalyst (around 15,000 kg).

- Low Temperature (335°C - Solid blue line): At the lowest temperature, the reaction is extremely slow. The curve is much shallower, and more catalysts (over 45,000 kg) is required to attain even close to the 99% conversion point.
- The "DESIGN" Point (355°C - Orange dash-dot line): This plot shows the design temperature selected for this design of a reactor. It is a balance between the extreme temperatures, and it reaches the desired 99% conversion (represented by the horizontal yellow dashed line) at a final catalyst weight of about 44,610 kg.
- Influence of Pressure Drop: Although the curves are typical for thermodynamic curves, the flattening of the curves for the higher weights of the catalyst, especially for lower temperatures, is influenced by the pressure drop. As the length of the catalyst bed increases, so does the pressure drop, resulting in a decrease in partial pressures of the reactants and a further reduction in the rate of reaction, which is already slow and is required to reach a 99% conversion.

The Effect of Catalyst Weight on Conversion of Our Chosen Temperature $T = 355\text{ C}$

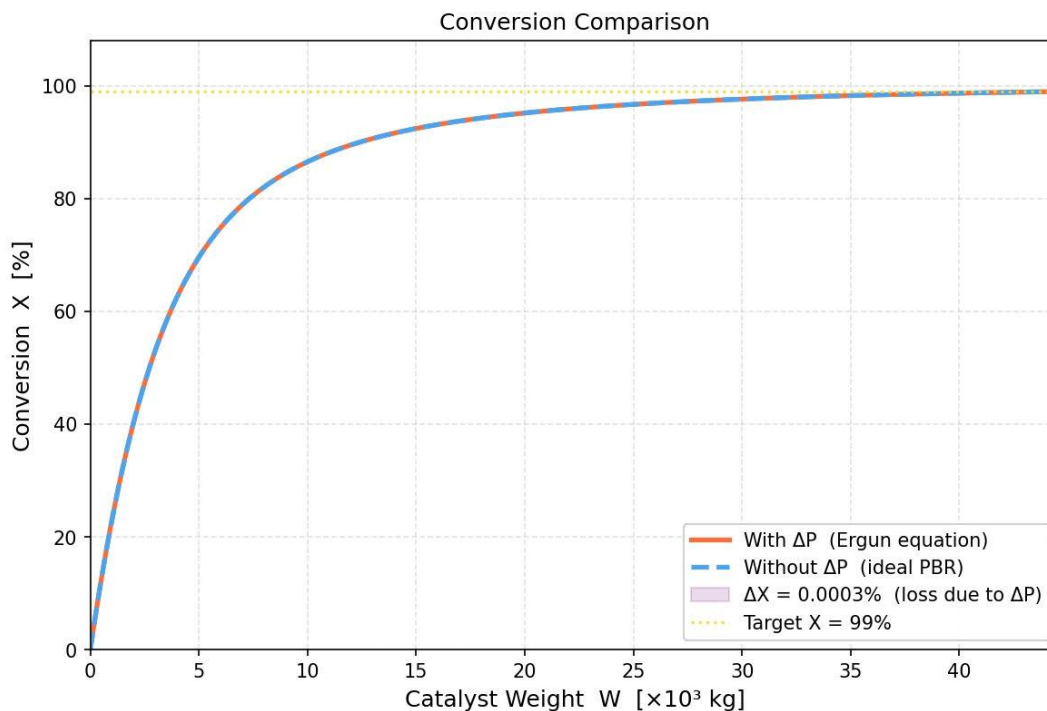


Figure 2: The Effect Catalyst Weight on Conversion of Our Chosen Temperature

This figure provides a comparison chart for conversion (X) as it relates to the weight (W) of the catalyst in the packed bed reactor system. The x-axis shows the weight of the catalyst in units of $\times 10^3$ kg, while the y-axis shows the percentage conversion. The conversion is the measure of the reaction from the reaction to the product as the reaction progresses in the reactor. At the initial condition, when the weight is 0, the conversion is 0 because there is no catalyst and hence no reaction.

There are two major types of curves shown in the figure. One is the curve representing the scenario when there is pressure drop (ΔP), shown as a solid line. The pressure drop is determined using the Ergun equation. The other curve, shown as a dashed line, is the ideal scenario when there is no pressure drop. By looking at the two curves, they are almost identical, indicating that the effect of the pressure drop is negligible.

To begin with, both curves show a sharp increase, which implies that the rate of reaction is high and that conversion is rising rapidly with a small amount of catalyst added to the system. After this, there is a gradual progression towards a plateau, which implies that with an increase in the amount of catalyst, there is still an increase in conversion, though with diminishing returns due to a reduction in reactant concentration and a concomitant reduction in reaction rate.

There is a shaded region between the curves, showing the difference in conversion due to pressure drop, which is calculated to be $\Delta X = 0.0003\%$. This is an extremely low figure, suggesting that there is no effect of pressure drop on the reactor's performance, and practically, one could say that this is a near-ideal condition for a packed bed reactor (PBR).

A horizontal line is a plot of 99% conversion, which is the required conversion level in this system. Both curves reach this level of conversion at almost the same weight as catalysts, suggesting that there is no effect of pressure drop on this system's design requirements.

Conversion Versus Reactor Volume

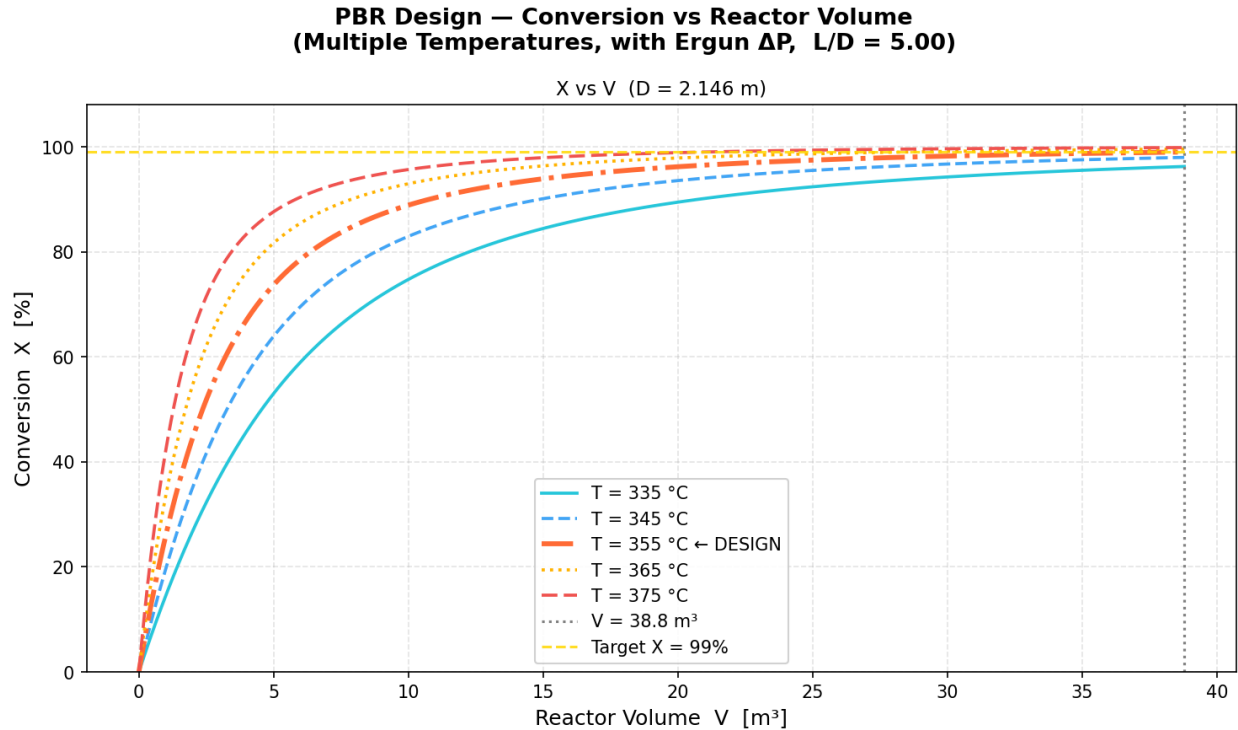


Figure 3: Conversion Versus Reactor Volume

The figure represents the relationship between conversion, denoted as X , and volume, denoted as V , for a PBR at different temperatures. The x-axis represents volume in cubic meters, denoted as m^3 , while the y-axis represents conversion as a percentage. Conversion refers to the amount of reactant converted into a product. At the origin, where volume equals zero, conversion is also zero, as there is no reaction occurring in the system.

The figure displays several curves, each representing a reaction at a different temperature, ranging from 335°C to 375°C . From the figure, higher temperatures result in higher conversion at a faster rate. For example, at 375°C , the reaction exhibits a sharp increase in conversion at a low volume, while at 335°C , the reaction exhibits a slow increase in conversion at a higher volume.

Another notable aspect of the figure is that all the curves exhibit a sharp increase at first, followed by a steady-state region where the reaction rate is low, as the amount of reactions is low. As a result, the curves tend to a certain limit, where conversion approaches 100%. However, the reaction rate is high at first, but as the reaction progresses, the number of reactions available reduces, leading to a low reaction rate.

The target conversion is also represented on the graph, where a horizontal line at 99% indicates the target conversion. A vertical line at 38.8 m³ represents the design volume, where at the design temperature of 355°C, the system can achieve almost 99% conversion, indicating that the design volume is suitable for the reaction to achieve 99% conversion.

3.2 Multi-Tube Configuration Analysis

3.2.1 Tube Length Affecting Pressure Drop for different N_t

Impact of the Number of Tubes (N_t)

As can be seen, the results show a dramatic inverse relationship between the number of tubes and the total pressure drop (ΔP).

- **High Pressure Drop ($N_t = 1000$):**
Since the number of tubes is low, the total mass flow rate is forced to pass through a small area, resulting in a high mass flux (G). The Ergun Equation indicates that the pressure drop is directly proportional to the square of the mass flux (G^2).
- **Reduction of Pressure Drop:**
Increasing the number of tubes from 1000 to 8000 reduces the pressure drop to almost zero. By increasing the number of tubes, the gas velocity in individual tubes is reduced, resulting in the minimization of frictional losses on the catalyst bed.

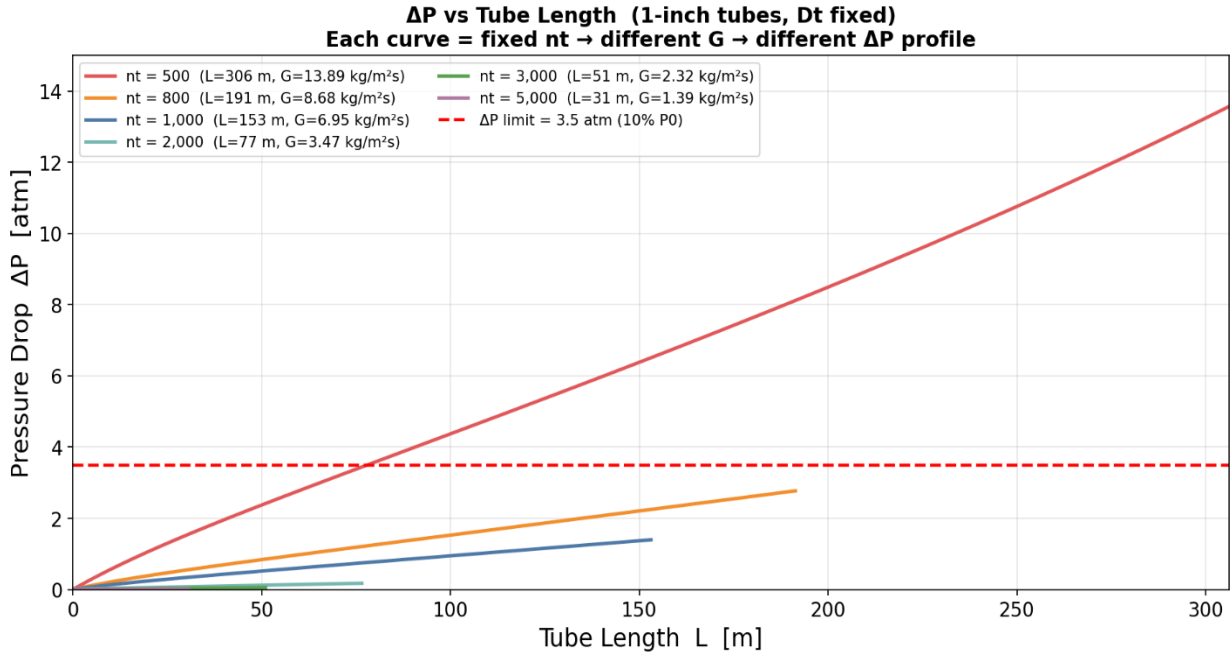


Figure 4: Pressure Drop Versus Tube Length Curve

Relationship between ΔP and Tube Length (L)

The graph indicates that pressure drop increases linearly with tube length for all cases.

- **Linear Accumulation:** As gas moves down the tube, there is an accumulation of catalyst particles. This results in a loss of momentum.
- **The 'Cut-off' Phenomenon:** It is observed from the graph that for $N_t = 1000$, the line reaches a higher length (approximately 154 m) compared to $N_t = 8000$ (approximately 20 m). This is due to the fact that to maintain a constant reactor volume of 38.8 m³, for a lesser number of tubes, the length of the tube would be greater.
- **Choking Hazard:** The slope for $N_t = 1000$ is very steep. This indicates a 'choking' hazard. If there is an increase in pressure drop by more than 10% of the original pressure ($y < 0.9$), there would be a considerable effect on conversion due to gas expansion.

Design Optimization and Trade-offs

The analysis shows that the main trade-off in the design optimization for a multi-tubular PBR is:

- Large N_t (Preferred for ΔP):
The use of 4000 to 8000 tubes keeps the pressure drop well within safety limits (below 25 kPa). This keeps the concentration of the reactants high, thereby keeping the rate high.
- Small N_t (Economic Constraint):
Though it is economically favorable to have fewer tubes to minimize the cost of tubes and catalyst used for filling the tubes, ΔP is so high that the required compression work is impractical.

In the case of the Cumene production reactor, a configuration where $N_t \geq 4000$ is recommended. This configuration ensures that the pressure drop is kept low enough for the required conversion of 99% to be achieved without making the tube too long.

3.2.2 Tube Size Affecting Pressure Drop for Different Tube Sizes

The influence of tube diameter on pressure drop was investigated for different numbers of tubes, keeping the volume of the reactor and the weight of catalyst constant. The results clearly indicate that tube diameter plays an important role in pressure drop, especially through its influence on tube velocity and mass flux. As the tube diameters increase, the cross-sectional area of each tube also increases. This, in turn, decreases the superficial velocity and mass flux, which subsequently decreases the frictional resistance. Therefore, as the tube diameters increase, the pressure drop also decreases, as predicted in the Ergun equation.

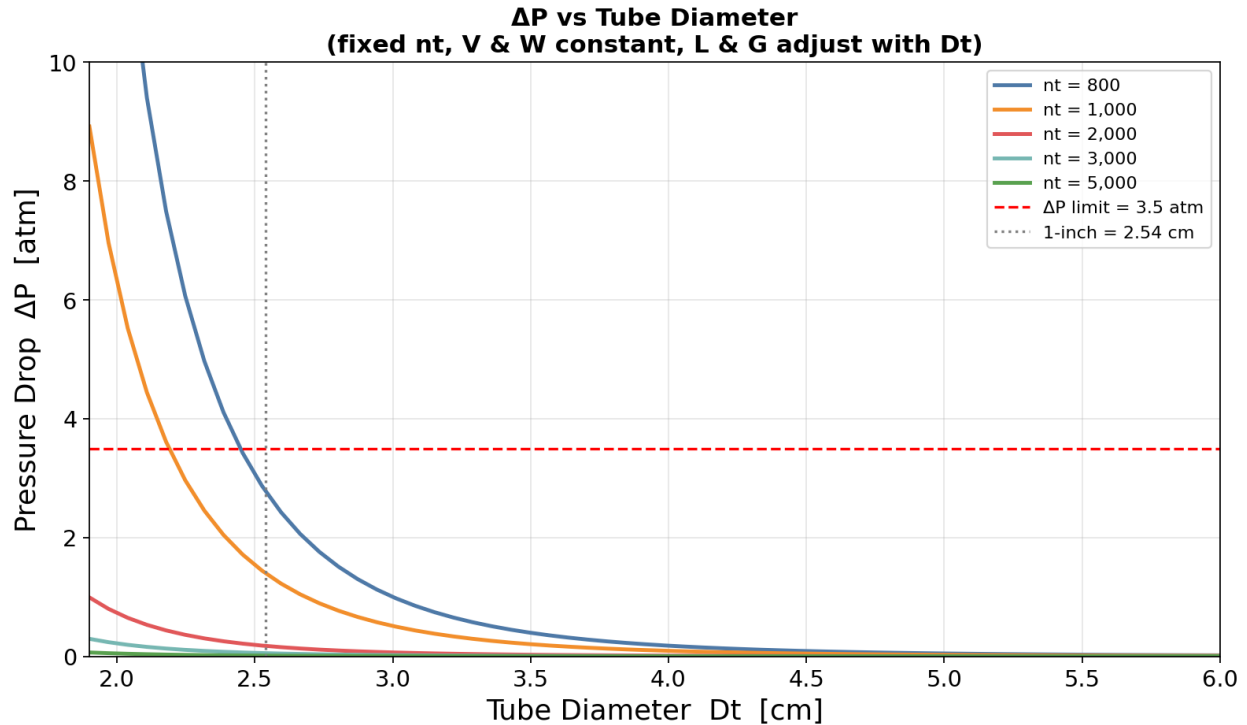


Figure 5: Pressure Drop Versus Tube Diameter for different N_t

Key observations:

- Pressure drops decrease rapidly as tube diameter increases, especially in the small diameter range.
- For tube diameters below about 2.5 cm, the pressure drop is increased sharply. In configurations with fewer tubes, $N_t < 1000$ tubes, it is observed that ΔP exceeds the maximum permissible value of 3.5 atm, making such configurations impractical due to increased resistance.
- The increase in the number of tubes also decreases the flow rate per tube, which in turn decreases the velocity and mass flux. Therefore, increasing the tube numbers, $N_t > 2000$, keep pressure dropping much below the allowable value (3.5 atm) for all diameters.
- As can be observed, the pressure drop at the reference diameter is near the allowable limit for lower tube counts, while for higher tube counts, the pressure drops stay within safe limits. This again emphasizes the importance of the choice of tube diameter as well as the number of tubes.

Design Trade-Off

Smaller tube diameters enhance the heat transfer performance but cause a considerable increase in pressure drop. On the other hand, larger tube diameters can cause a decrease in pressure drop but may influence the compactness and performance of the reactor.

Overall, the results clearly show that the tube diameter is a very critical design parameter with a highly nonlinear effect on pressure drop. Optimizing the tube diameters, as well as the number of tubes, is very critical for feasible operation within reasonable pressure drop limits.

3.2.3 Length Affecting Pressure for Different Particle Sizes

An analysis of the effect of the size of the catalyst particles on pressure drop (ΔP) was performed as a function of tube length, maintaining a constant reactor volume, and using 1-inch tubes. The results showed that the size of the particles has a major impact on pressure drop, as this factor is directly related to flow resistance.

As the gas passes through the catalyst bed, smaller catalysts result in narrower channels and hence an increased frictional resistance. However, larger catalysts result in larger free spaces and hence an easier flow of gas. This is in accordance with the Ergun equation, where pressure drop is inversely proportional to catalyst diameter.

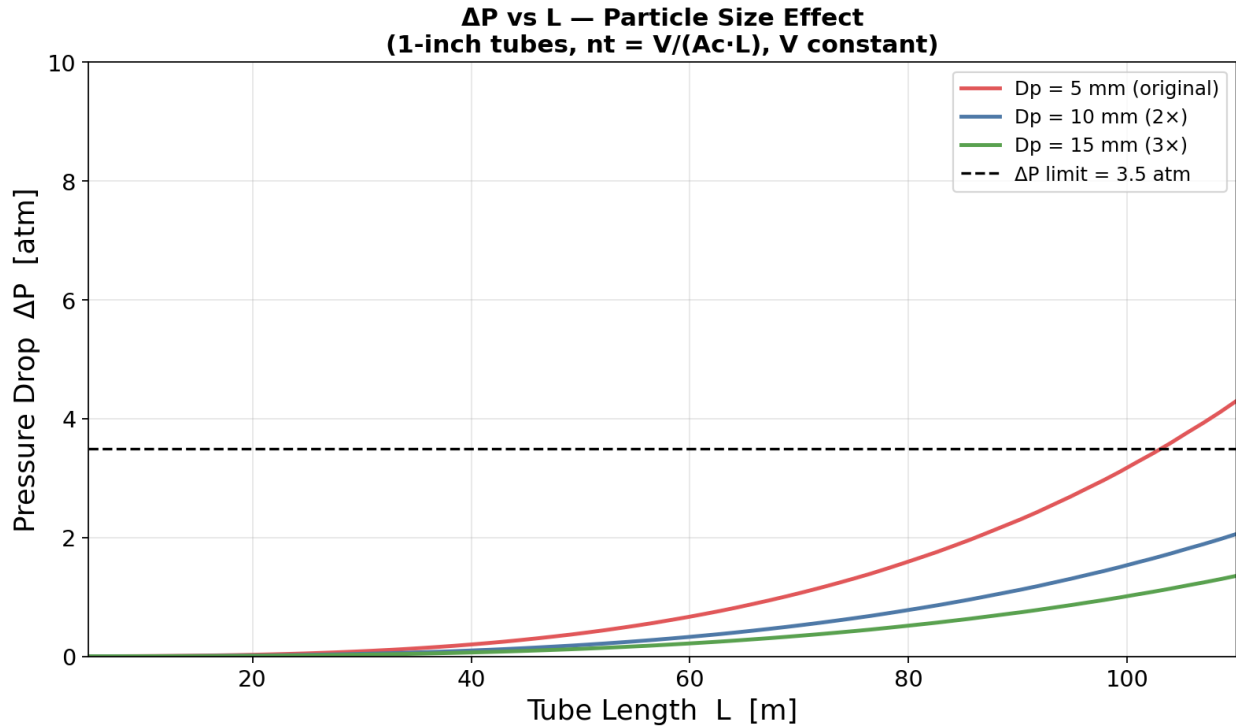


Figure 6: Pressure Drop Versus Tube Length for different D_p

Key Observations:

- For all particle sizes, ΔP increases as the tube length increases. This is due to cumulative frictional losses along the packed bed as the fluid travels a longer distance.
- Smaller particles, ($D_p = 5$ mm), produce a higher pressure drop compared to larger particles, ($D_p = 10$ mm, 15 mm). This is due to the increased resistance offered by smaller particles.
- The increase in pressure drop is nonlinear for tube length for smaller sizes of the particles. This reveals the fact that the resistance effects become more pronounced as the flow progresses along the tube.
- In the case of the smallest particle size (5 mm), the pressure drop is found to be above the permissible limit of 3.5 atm for longer tube lengths, i.e., around 100 m. This means that these conditions may not be feasible for operation. However, for larger particle sizes, they lie within the limits for the entire range.

Design Trade-off

Although larger particle size results in a decrease in pressure drop, the catalytic surface area and rate of reaction may be decreased as well. Thus, an optimum particle size must be found for a given situation.

Overall, the results emphasize the importance of particle size effect on pressure drop. The particle size must be carefully selected for the reactor operation to be within acceptable pressure limits, with efficient performance in the reaction process.

3.3: Sensitivity Analysis

3.3.1 Varying Inlet Pressure Affecting Pressure Drop

3.3.2 Sensitivity of Pressure Drop to Catalyst Particle Size

Comparative Overview of Tube Configurations ($N_t = 2000$ vs. $N_t = 4000$)

A comparative analysis was carried out for the two reactor configurations to assess the level of sensitivity for the different operational and physical parameters involved in the process. This comparison is visualized through Figure 1 ($N_t = 4000$ tubes) and Figure 2 ($N_t = 2000$ tubes), which highlight how design choices impact reactor stability.

- $N_t = 2000$ (Longer Tubes): With fewer tubes, the length of the tubes must be increased, which in this case was approximately 77 meters in length. This configuration was found to be highly sensitive, as the pressure ratio, denoted as (y), was observed to decrease to a mere 0.91, even at a pressure of 10 bar, which is very close to the absolute design limit of $y = 0.9$.
- $N_t = 4000$ (Optimal Design): This configuration, where the number of tubes was doubled, resulted in a significant decrease in the length of the tubes, which was approximately 38 meters in length. This caused a significant decrease in the mass flux, denoted as (G), which in turn made the pressure profile highly stable, with (y) remaining above 0.99 even at lower pressure levels.

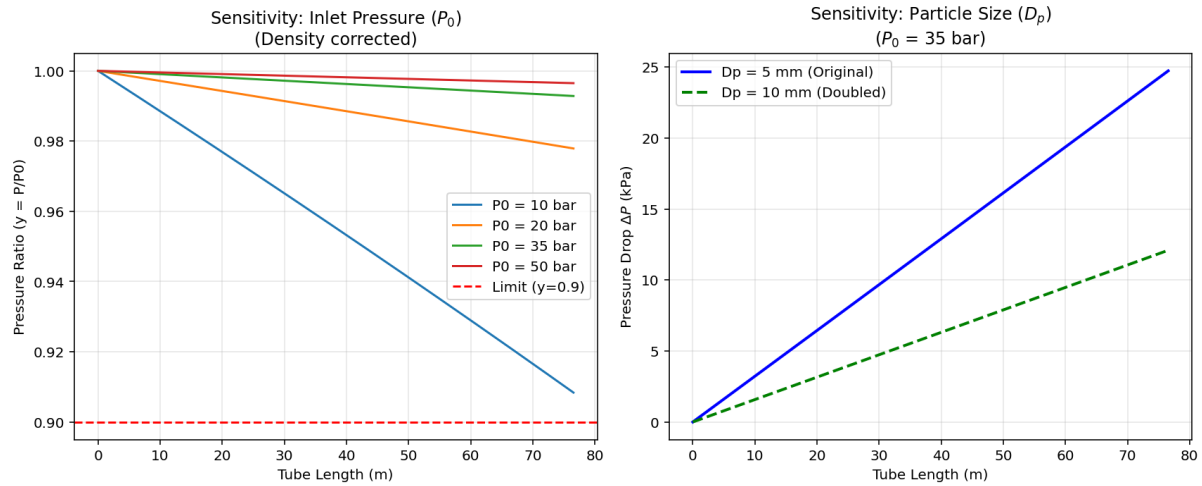


Figure 7: Sensitivity Analysis for the Marginal Reactor Configuration ($N_t = 2000$ tubes)

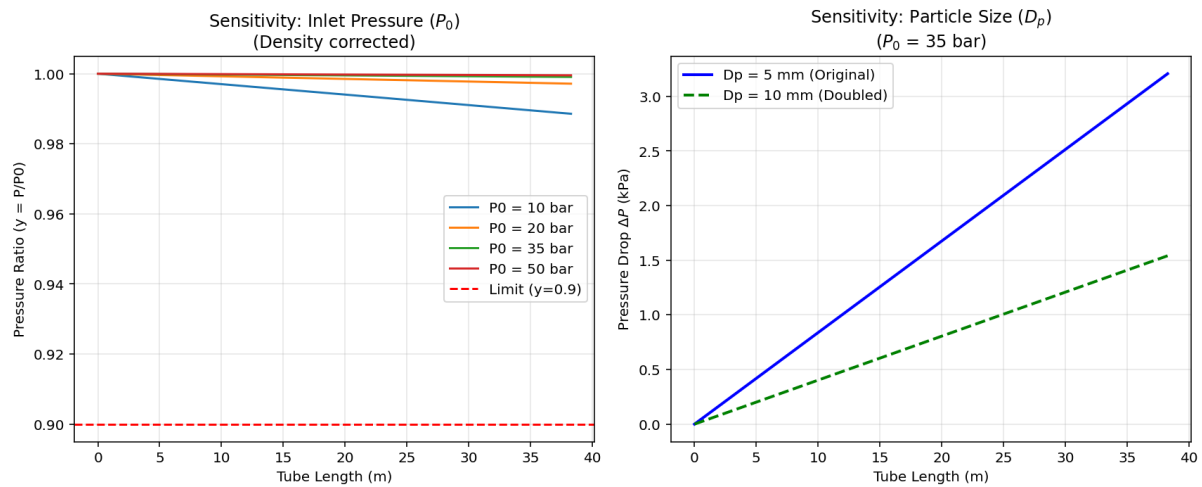


Figure 8: Sensitivity Analysis for the Optimized Reactor Design ($N_t = 4000$ tubes)

Sensitivity of Inlet Pressure to ΔP

As shown by the sensitivity analysis for the Inlet Pressure, the pressure drop is more critical for lower pressures.

Density Effect: When the inlet pressure increases from 10 bar to 50 bar, the gas density will increase. This will cause the volumetric flow rate and linear velocity of the gas to decrease. This will cause the frictional resistance to decrease.

Sensitivity of Catalyst Particle Size to ΔP

As required by the design specifications, the catalyst particle size was doubled from 5 mm to 10 mm to assess the impact on the momentum balance.

Significant Reduction in ΔP

As shown in the $N_t = 2000$ configuration, the doubling of the catalyst size resulted in a reduction of the maximum pressure drop from 25 kPa to 12 kPa.

Trade-off

As shown by the $D_p = 10$ mm configuration, larger catalyst sizes effectively halve the pressure drop by providing larger void channels for the gas flow. However, it should be noted that larger catalyst sizes usually provide a smaller surface area for the reaction. Nevertheless, from a momentum balance point of view, larger catalyst sizes are extremely favorable for maintaining $y > 0.9$ for designs with fewer tubes.

3.3.3 Bed Porosity Affect Pressure Drop

This figure shows the relationship between the void fraction (ϕ) and the pressure drop (ΔP) in a packed bed reactor. The horizontal axis indicates the void fraction, which is the space between catalyst particles in a reactor. The vertical axis indicates the pressure drop-in units of atm. The pressure drop is the loss in pressure that occurs when fluid moves through a packed bed reactor. This is a key consideration in a reactor design because a high pressure drop increases the cost of operation.

From the figure above, the pressure drops decrease with the void fraction. When the void fraction is relatively low (0.30), the pressure drop is high at about 16 atm. This is because the particles are packed tightly in the reactor, creating a smaller space for the fluid to flow through. As a result, the fluid experiences a high-pressure drop. When the void fraction increases, the particles become spaced out, making it easy for the fluid to flow through the reactor; therefore, the pressure drop decreases.

This curve has a sharp drop in the initial stages, followed by a gradual decrease. This implies that an increase in the value of ϕ from 0.30 to 0.40 results in a substantial decrease in ΔP , but any increase in the value of ϕ beyond this point will only provide diminishing returns. For example, an

increase from 0.50 to 0.60 will again result in a decrease in ΔP , but this will be less substantial compared to the earlier range.

A vertical dashed line is used for the design value of the void fraction, which is equal to 0.5. This is the point where the pressure drop is equal to 2.7 atm. A horizontal dotted line is used for the 10% of the inlet pressure (P_0), which is equal to 3.5 atm.

From this analysis, for the design point where the value of the void fraction is 0.5, the pressure drop is within the acceptable limits. However, void fractions that are very low, i.e., lower than 0.45, will result in a pressure drop that exceeds the acceptable limits.

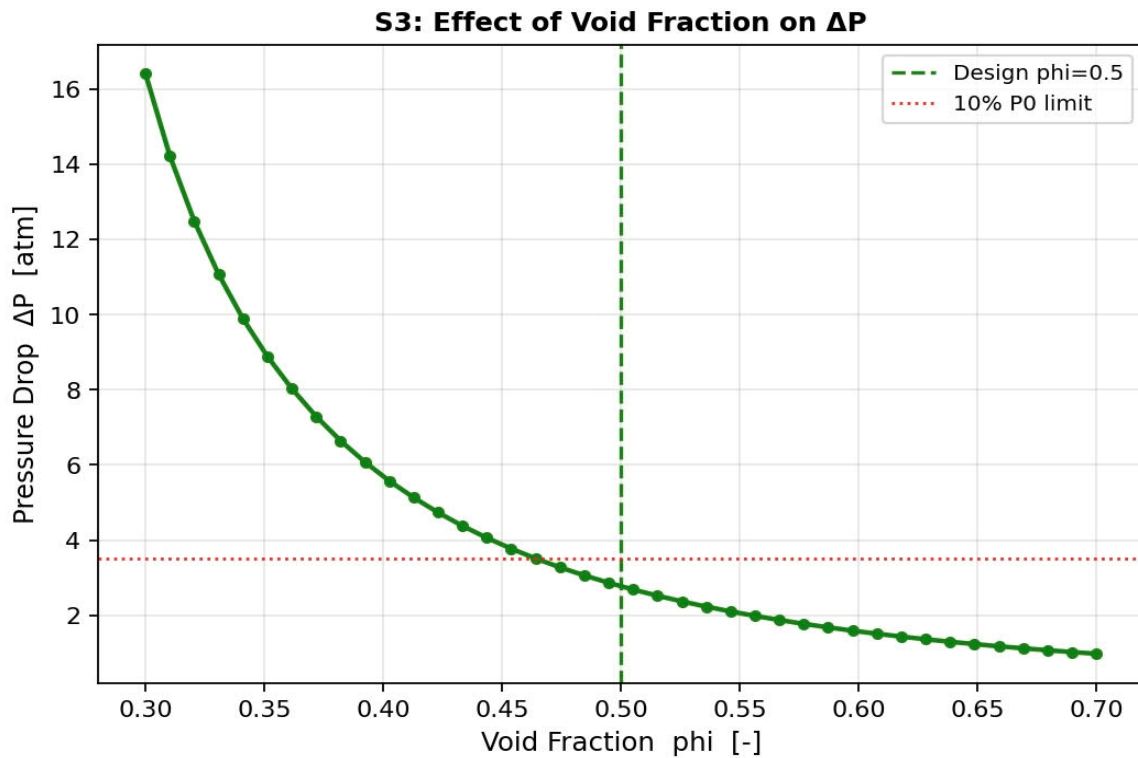


Figure 9: Pressure Drop Versus Void Fraction

3.3.4 Sensitivity of L/D ratio to Pressure Drop

This figure illustrates the relationship between the pressure drop (ΔP) and the length-to-diameter ratio (L/D) for a packed bed reactor under constant volume. The x-axis represents the ratio of length to diameter, whereas the y-axis represents the pressure drop in atm using a logarithmic scale.

According to this graph, the pressure drops increase with increasing values of the length-to-diameter ratio. This happens due to the constant volume constraint. An increase in the length-to-diameter ratio indicates a longer reactor with a smaller diameter, resulting in a smaller cross-

sectional area. As a result, the velocity of the fluid increases, resulting in a higher-pressure drop. At a lower level of the length-to-diameter ratio, the pressure drop is lower. As the value of the length-to-diameter ratio increases, the pressure increases rapidly. The curve indicates a smooth increase in pressure drop with increasing values of the length-to-diameter ratio. This indicates a high influence of geometry on pressure drop. A high value of the length-to-diameter ratio results in a high-pressure drop. In a reactor, a high-pressure drop is undesirable. Therefore, this aspect should be considered seriously. A vertical line at a value of 5 for the length-to-diameter ratio represents the chosen point. Along with this, the pressure drop is very low. A horizontal line at a pressure drop of 3.5 atm represents the maximum pressure drop. This pressure drop should be within 10% of the inlet pressure. According to this graph, even for a value of the length-to-diameter ratio of 50, the pressure drop is within this limit.

From this graph, we conclude that the pressure drop does not influence the choice of the length-to-diameter ratio. A high value of the length-to-diameter ratio is not desirable since this results in a longer reactor.

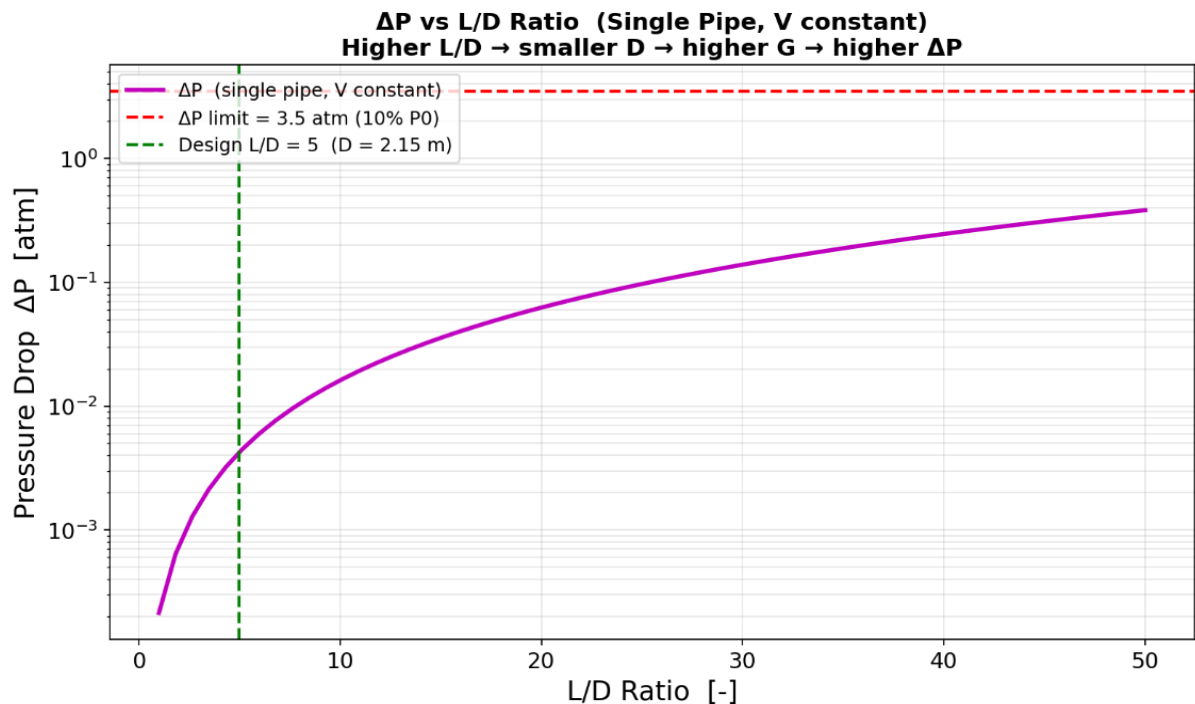


Figure 10: Pressure Drop Versus L/D Ratio

3.4 Polymath Code

```
# MBE
d(X)/d(W) = -ra / FA0

# Rate Law
ra = -K * CA * CB

# Stoichiometry
CA = CA0 * (1 - X) / (1 + eps * X) * y
CB = CA0 * (thetaB - X) / (1 + eps * X) * y

# Kinetics
K = K0 * exp(-E / (R * T))

# Parameters
K0 = 2.8e7
thetaB = 1.053
E = 104174
R = 8.314
T = 628.15
yA0 = 0.475
CA0 = 0.33882
eps = yA0 * (-1)

# Initial & Final Conditions
W(0) = 0
X(0) = 0
W(f) = 44610

# Pressure Drop (Ergun-based)
d(y)/d(W) = -(alpha / (2 * y)) * (1 + eps * X)
y(0) = 1
alpha = (2 * beta0) / (Ac * P0 * densc * (1 - phi))
beta0 = (G * (1 - phi) * ((150 * (1 - phi) * vis / Dp) + 1.75 * G)) / (gc * dens0 * Dp * (phi ^ 3))

# Physical Properties
densc = 1150
dens0 = 44.4216
phi = 0.5
```

```

# Physical Properties
densc = 1150
dens0 = 44.4216
phi = 0.5

# Geometry
Nt = 1
Dt = 2.15
Dp = 0.005
P0 = 3546375
pi = 3.141592654
Ac = (pi/4) * (Dt ^ 2)
V = W / densc
L = V / Ac

# Flowrates (kmol/s)
FA0 = 27.75
FB0 = 30.75
FI0 = 1.46

# Molecular Weights
MA = 42.08
MB = 78.11
MI = 44.1

# Mass Flow and Velocity
massflow = (FA0 * MA + FB0 * MB + FI0 * MI) / 1000
G = massflow / Ac

# Fluid Properties
vis = 1.76e-5
gc = 1

# Pressure Relations
P = y * P0
deltaP = P0 - P

```